RNA-inferred sprouting added to the Xenium atlas

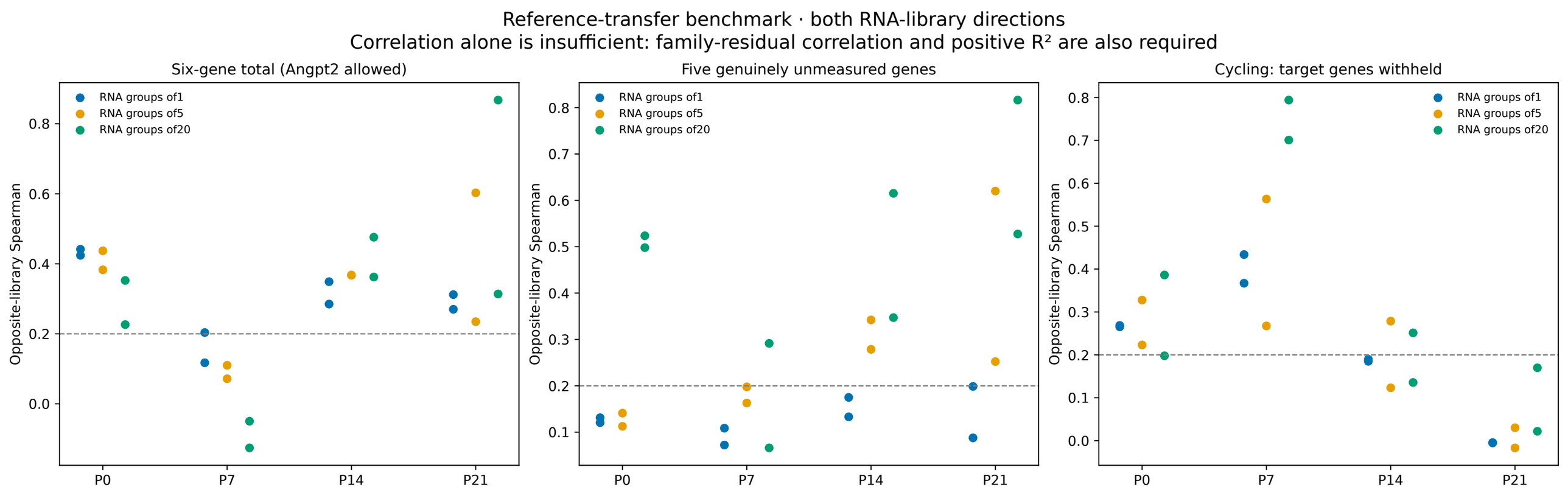
The new layer is a reference-derived estimate of sprouting-associated RNA, not an additional Xenium measurement. It annotates29,363 nuclear-clean previously labeled ECs across P0/P7/P14/P21. Every source cardiac profile remains present in the whole-heart annotation table; non-EC and excluded EC positions have no inferred score.

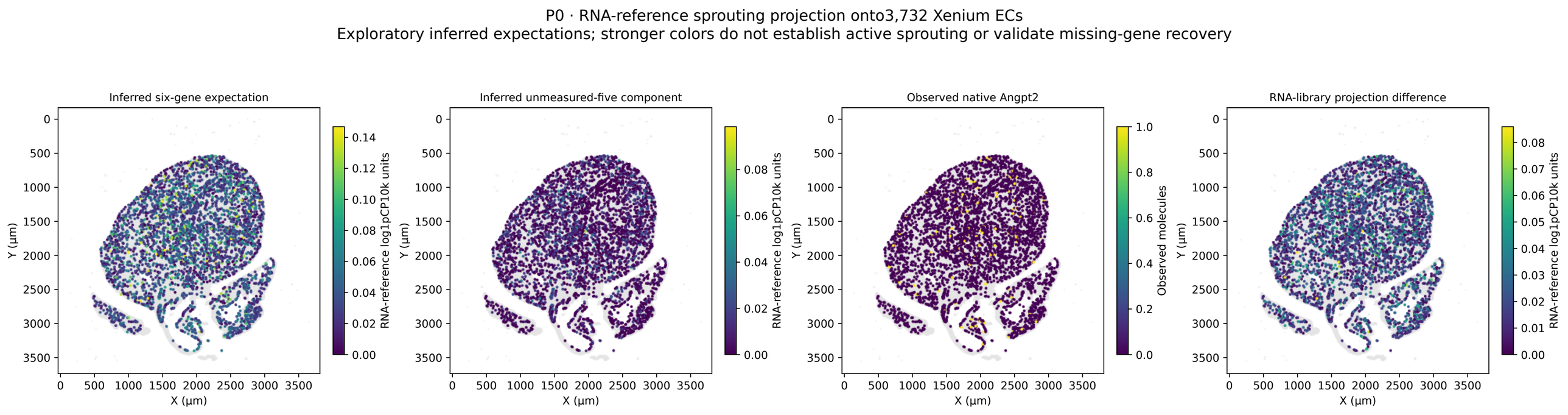
The RNA reference contains4,597 QC-selected endothelial nuclei. Same-age reference neighbors are selected with the shared measured genes. Their corrected RNA expression is transferred as a weighted expectation:30 cosine neighbors,temperature0.1, separate-assay scaling,SDfloor0.25 and clipping5. No sprouting label, desired score or physical coordinate selects neighbors.

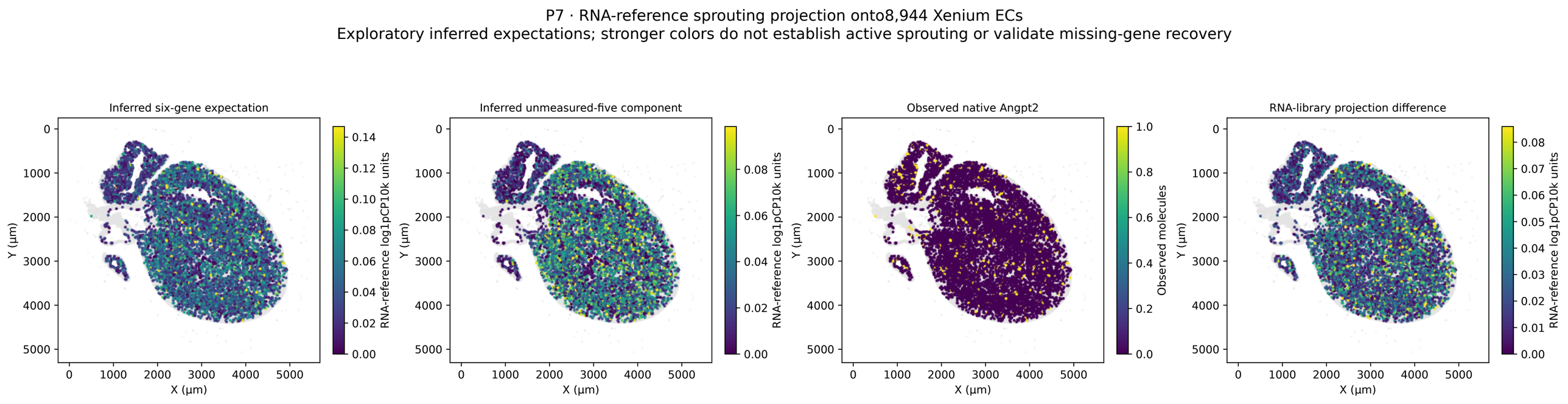
Two scores are supplied. The six-gene score averages inferred Apln,Esm1,Angpt2,Cxcr4,Pgf andAdm. The five-gene score excludes Angpt2 and exposes the genuinely unmeasured component. Measured Angpt2 counts remain separately available. Missing genes are never inserted into the observed403-gene count matrix.

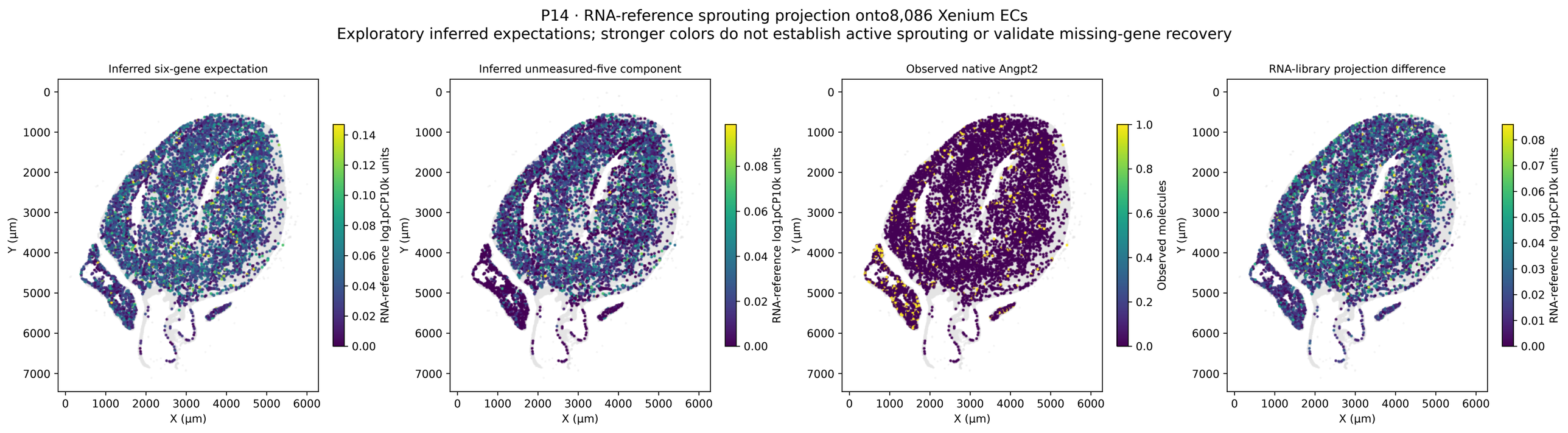
Individual scores,5-EC neighborhoods within75um and20-EC neighborhoods within150um are supplied. Neighborhoods are projected to disjoint, shared-gene-defined RNA metacells at matched target resolution. Pooled scores use a different reference resolution and should not be compared numerically with single-cell scores as if they were the same estimator.

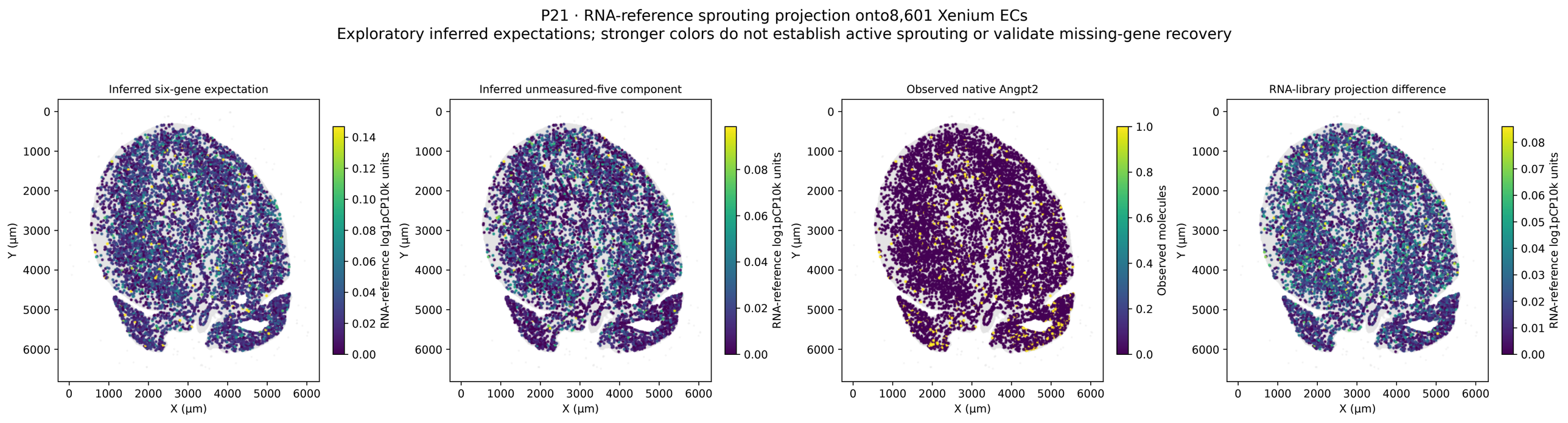
All spatial output is explicitly exploratory. The opposite-library RNA test supports parts of the total score, but recovery of the five unmeasured genes is weak at individual-cell resolution. Coarse neighborhood support is limited and age-dependent; it is not independent spatial validation or proof of new-vessel growth.


Reference-transfer benchmark · both RNA-library directions
Correlation alone is insufficient: family-residual correlation and positive R² are also required
Six-gene total (Angpt2 allowed)
Five genuinely unmeasured genes
Cycling: target genes withheld
Opposite-library Spearman
RNA groups of1
RNA groups of5
RNA groups of20
P0
P7
P14
P21


P0 · RNA-reference sprouting projection onto3,732 Xenium ECs
Exploratory inferred expectations; stronger colors do not establish active sprouting or validate missing-gene recovery
Inferred six-gene expectation
Inferred unmeasured-five component
Observed native Angpt2
RNA-library projection difference
Y (µm)
X (µm)
RNA-reference log1pCP10k units
Observed molecules
0
500
1000
1500
2000
2500
3000
3500
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.2
0.4
0.6
0.8
1.0
0.01
0.03
0.05
0.07

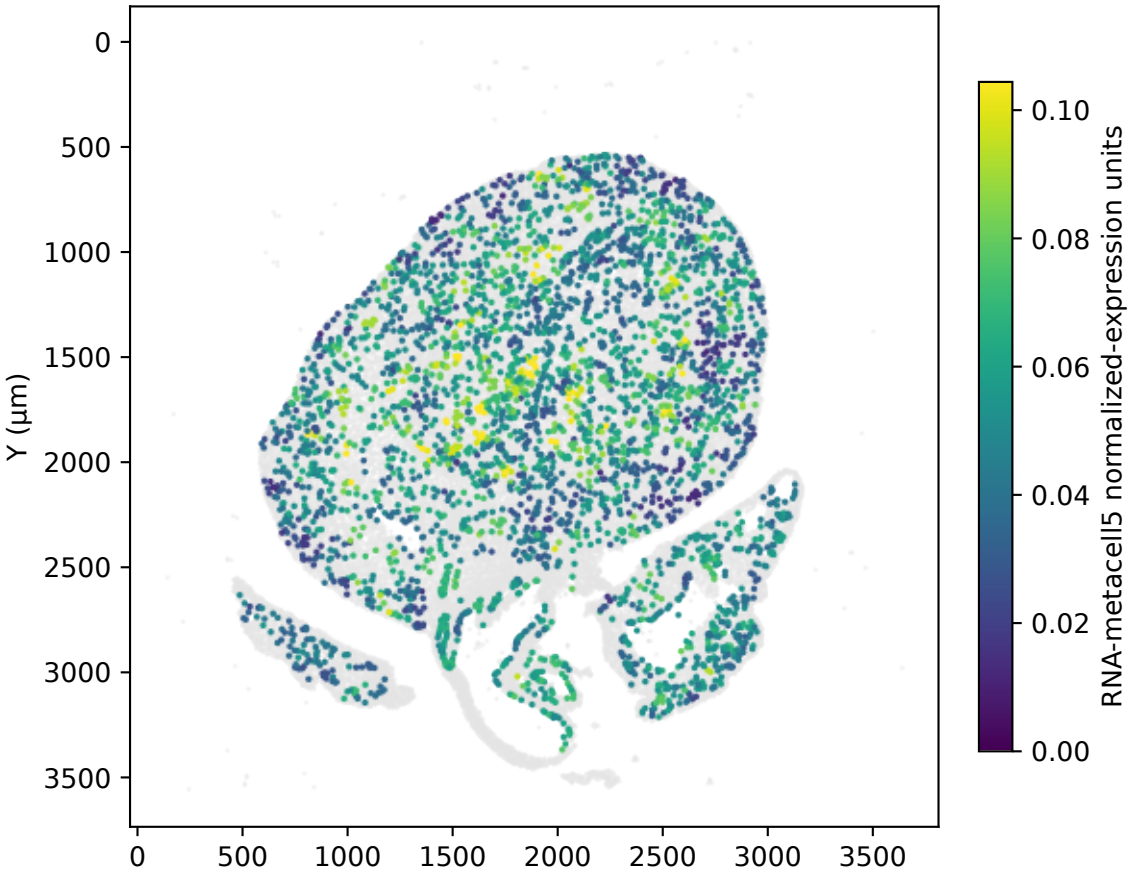

P7 · RNA-reference sprouting projection onto8,944 Xenium ECs
Exploratory inferred expectations; stronger colors do not establish active sprouting or validate missing-gene recovery
Inferred six-gene expectation
Inferred unmeasured-five component
Observed native Angpt2
RNA-library projection difference
Y (µm)
X (µm)
RNA-reference log1pCP10k units
Observed molecules
0
1000
2000
3000
4000
5000
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.0
0.2
0.4
0.6
0.8
1.0
0.01
0.03
0.05
0.07


P14 · RNA-reference sprouting projection onto8,086 Xenium ECs
Exploratory inferred expectations; stronger colors do not establish active sprouting or validate missing-gene recovery
Inferred six-gene expectation
Inferred unmeasured-five component
Observed native Angpt2
RNA-library projection difference
Y (µm)
X (µm)
RNA-reference log1pCP10k units
Observed molecules

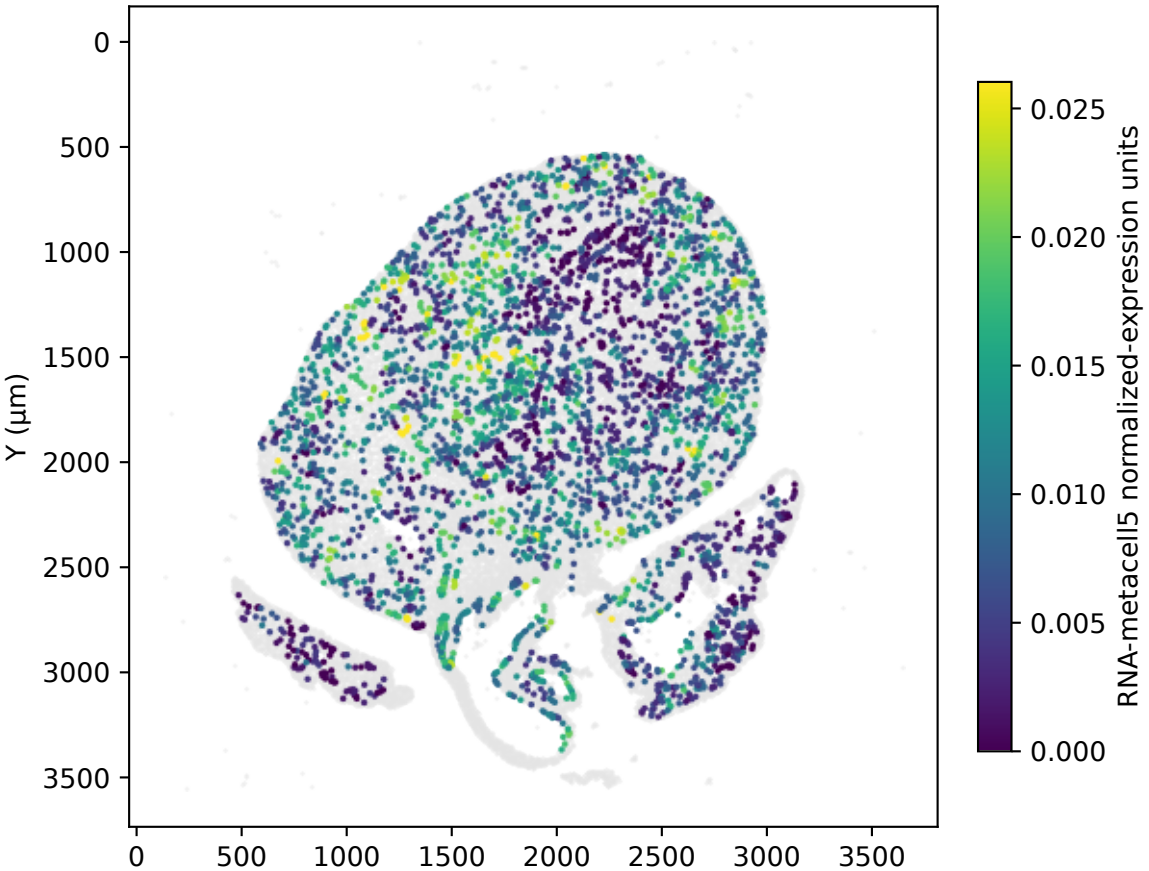

P21 · RNA-reference sprouting projection onto8,601 Xenium ECs
Exploratory inferred expectations; stronger colors do not establish active sprouting or validate missing-gene recovery
Inferred six-gene expectation
Inferred unmeasured-five component
Observed native Angpt2
RNA-library projection difference
Y (µm)
X (µm)
RNA-reference log1pCP10k units
Observed molecules
0
1000
2000
3000
4000
5000
6000
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.0
0.2
0.4
0.6
0.8
1.0
0.01
0.03
0.05
0.07

P0 · exploratory RNA-inferred neighborhood sprouting
Five ECs within75µm or20 within150µm; overlapping pools, not independent cells; scales differ by RNA reference resolution

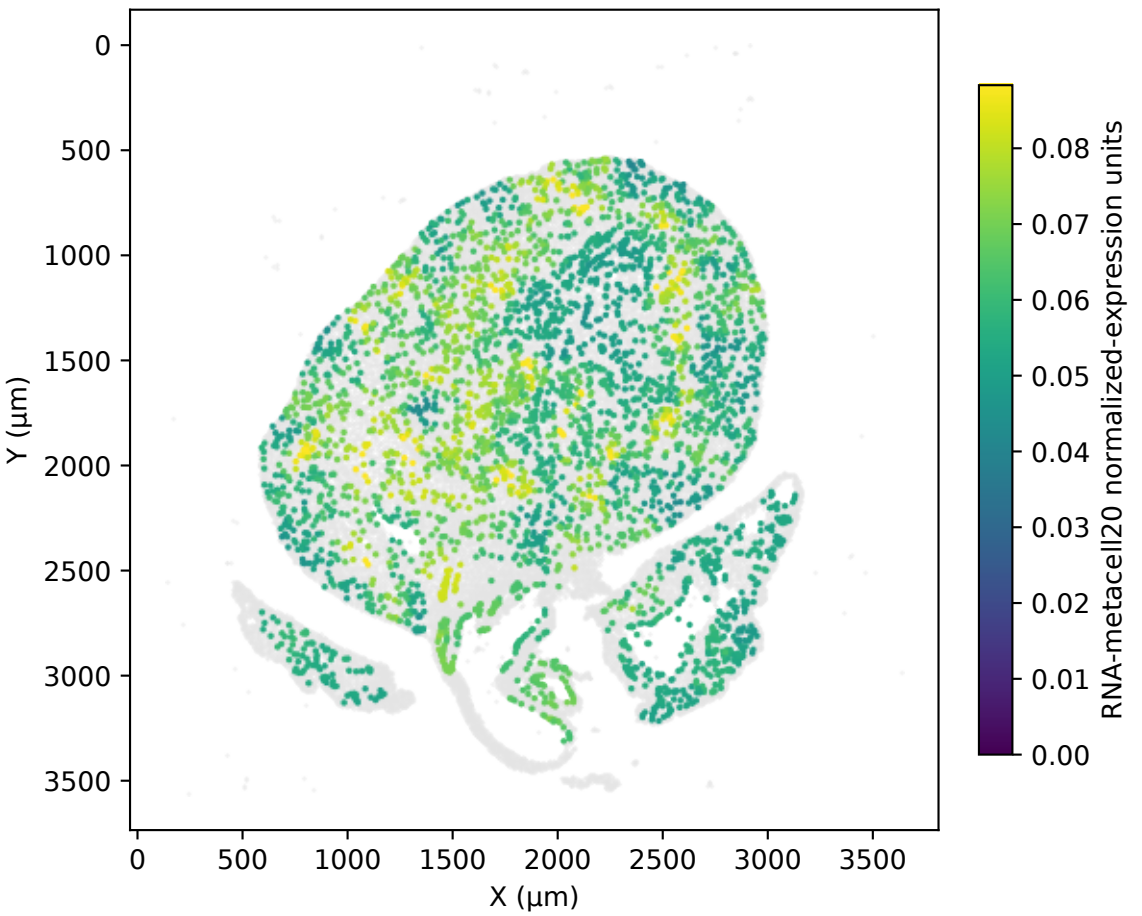
Six-gene expectation ·5 ECs



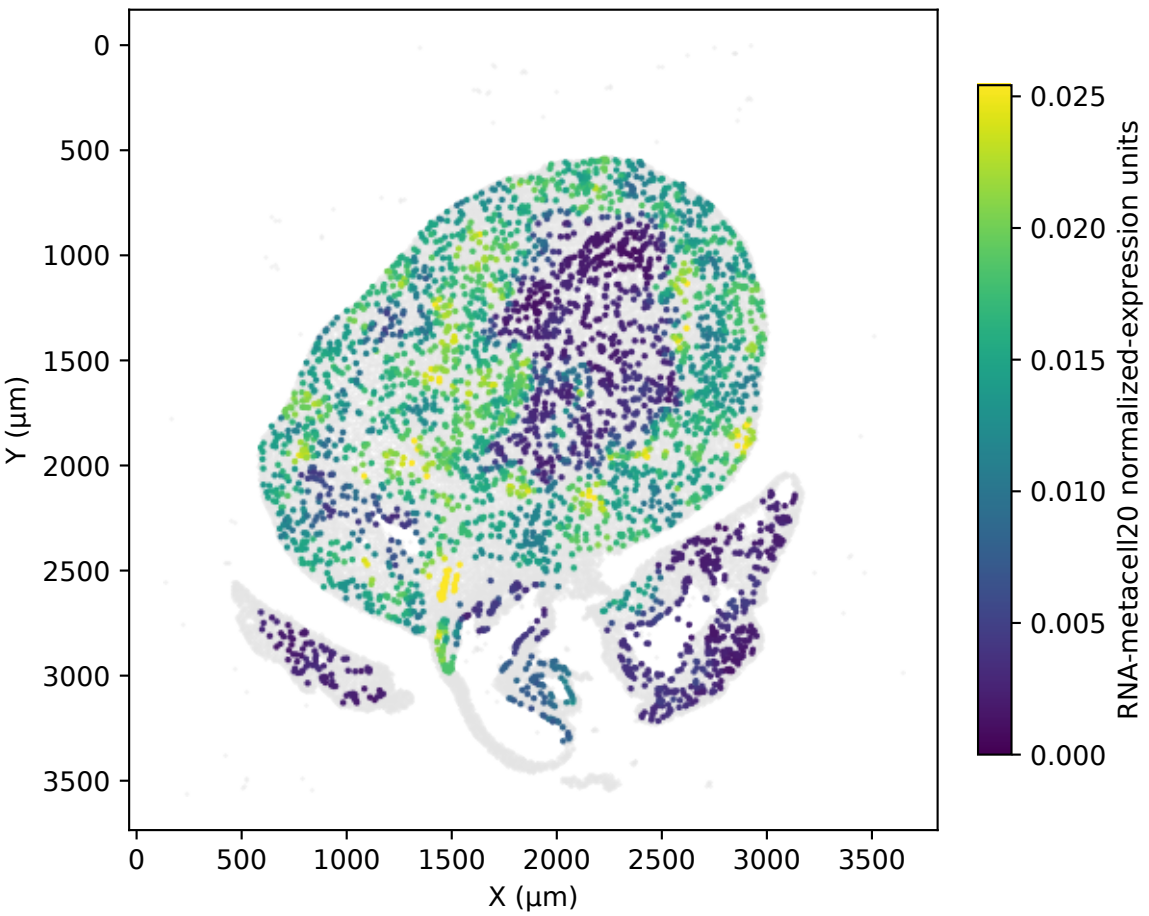
Unmeasured-five component ·5 ECs



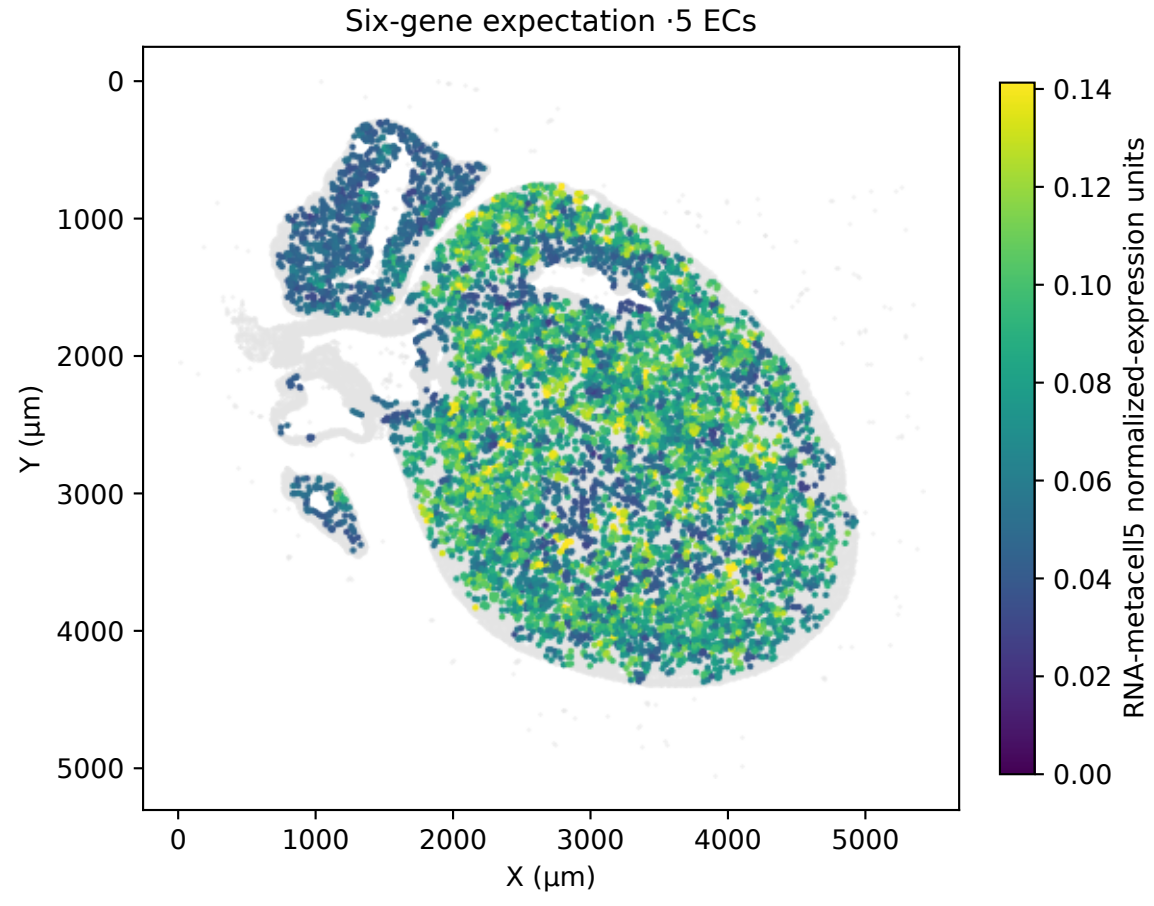
Six-gene expectation ·20 ECs

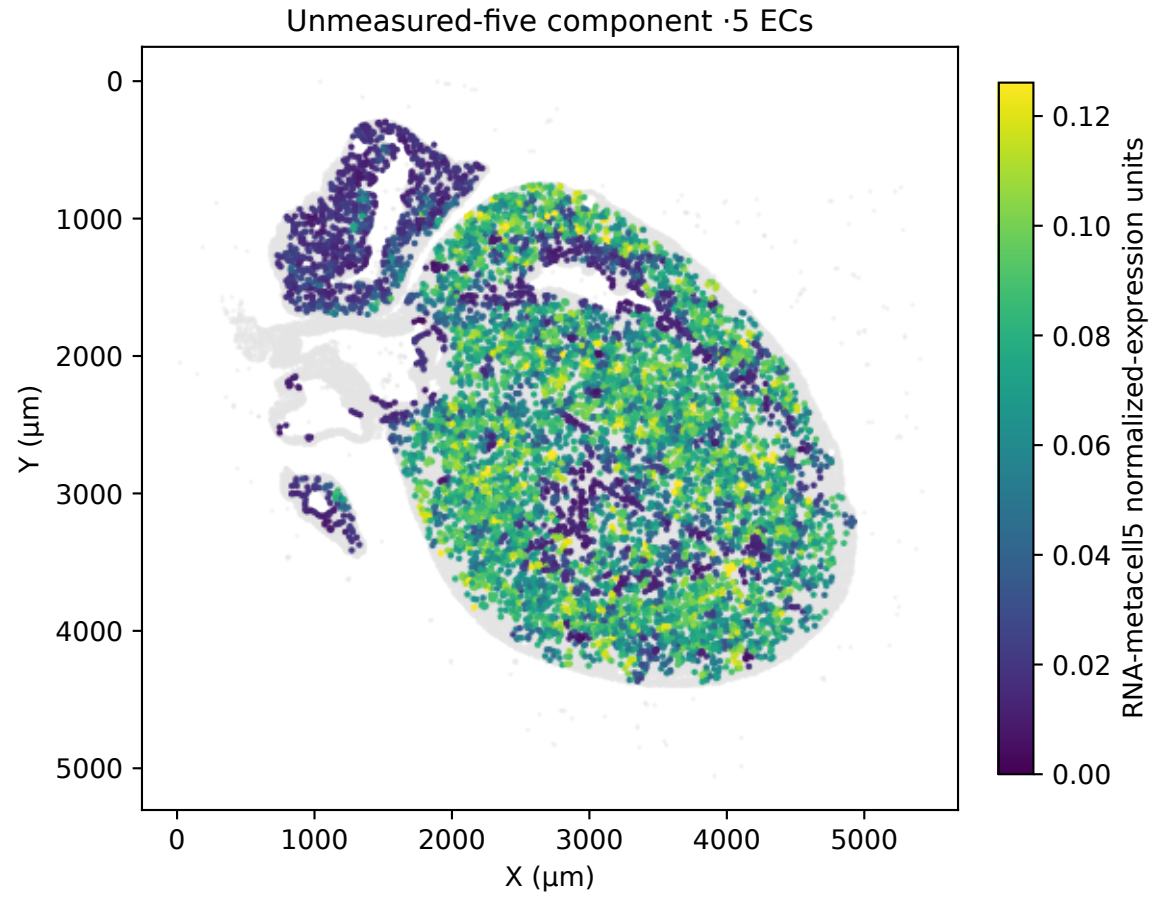


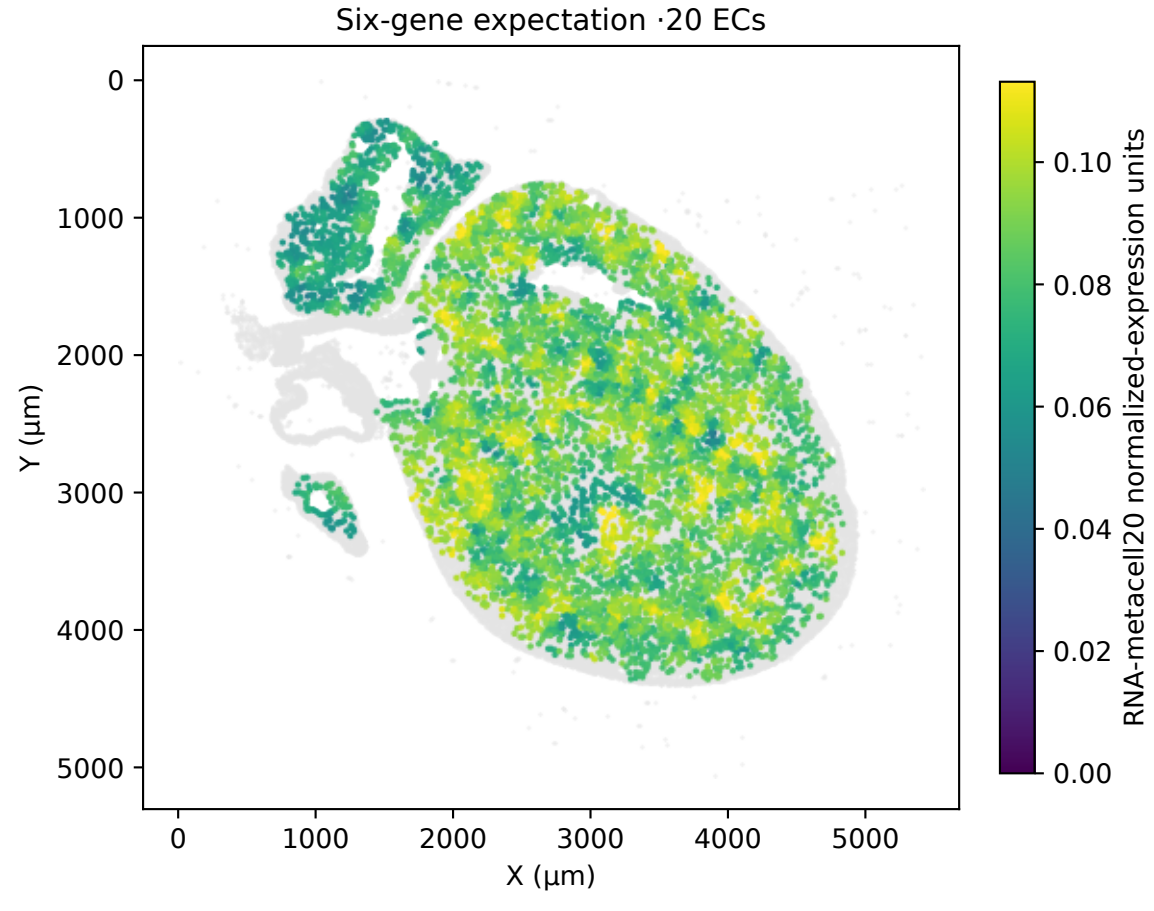
Unmeasured-five component ·20 ECs

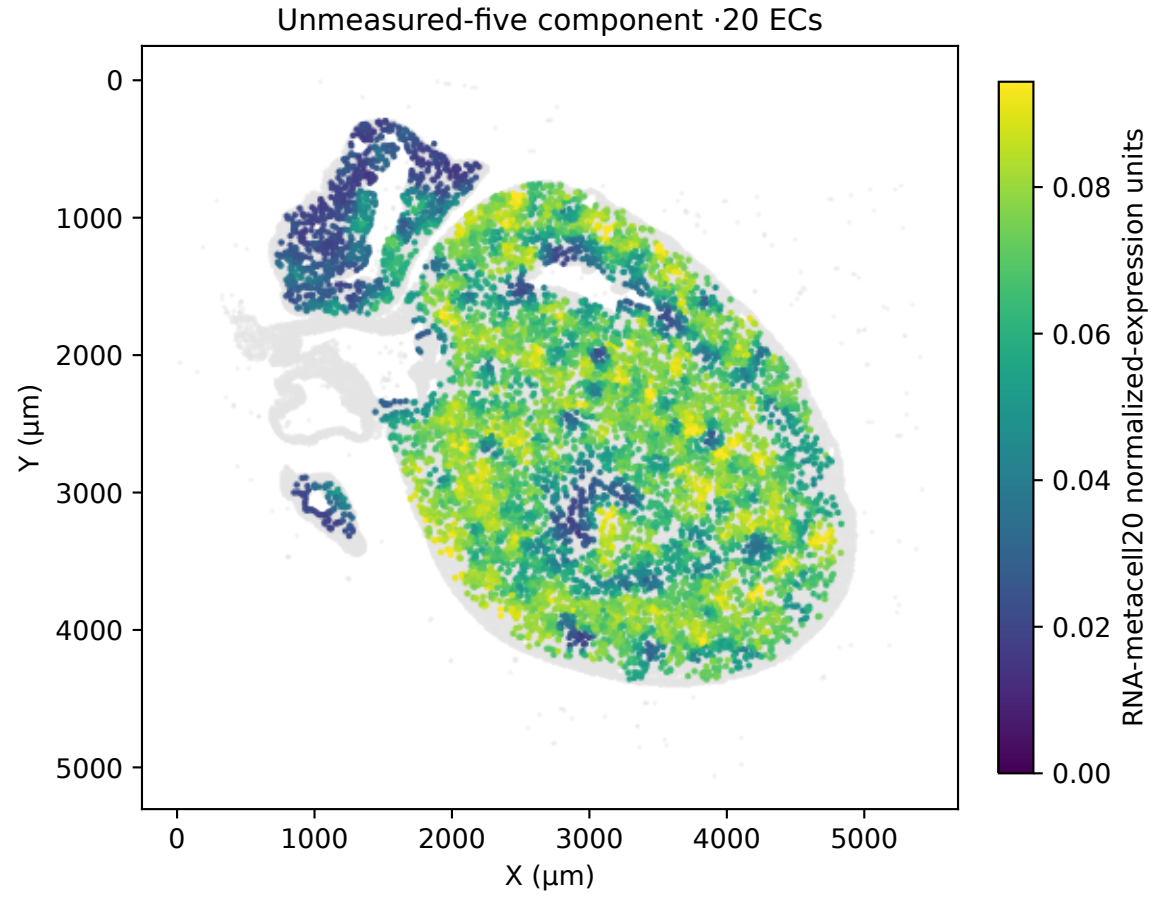


P7 · exploratory RNA-inferred neighborhood sprouting
Five ECs within75µm or20 within150µm; overlapping pools, not independent cells; scales differ by RNA reference resolution



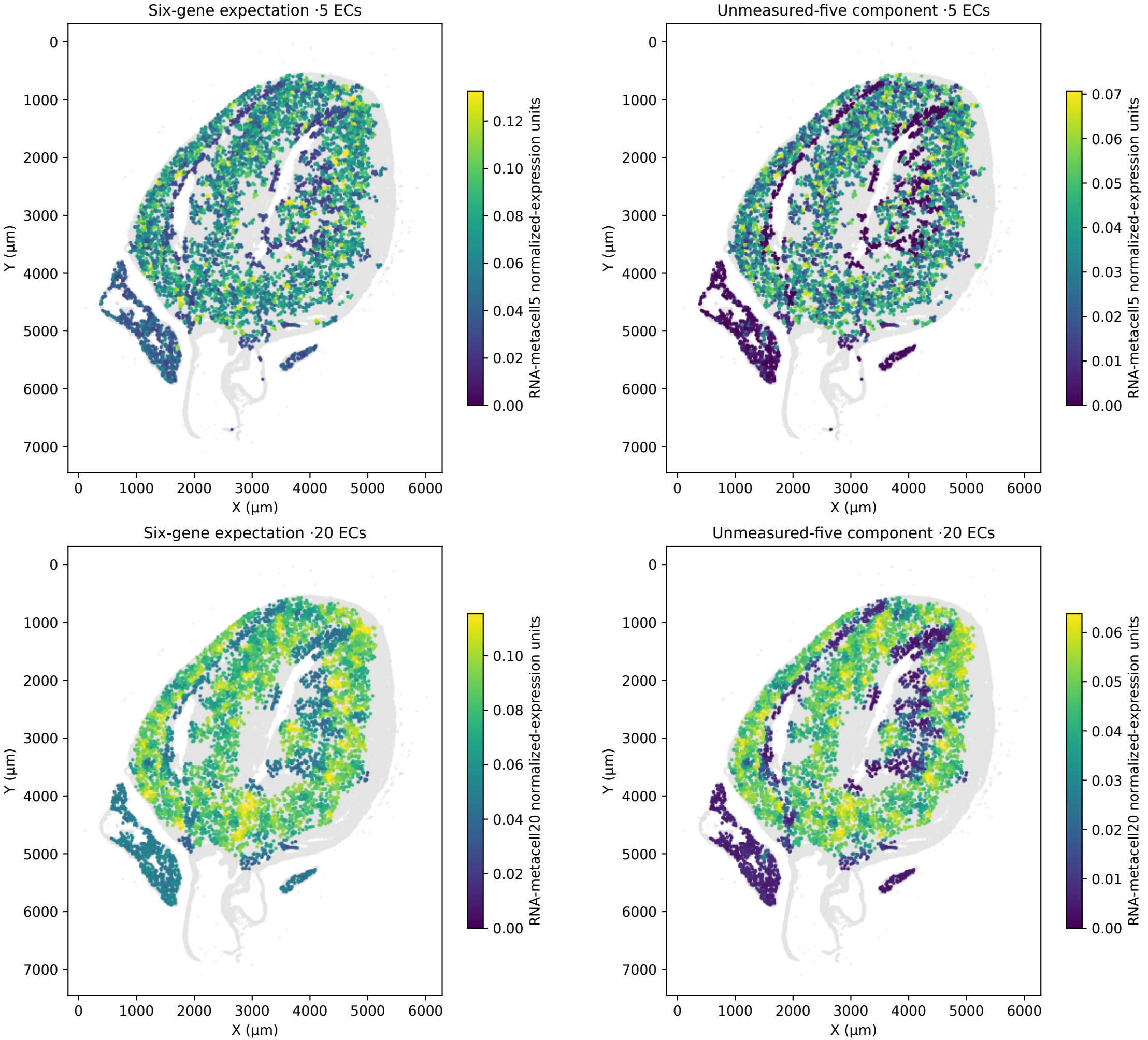






P14 · exploratory RNA-inferred neighborhood sprouting

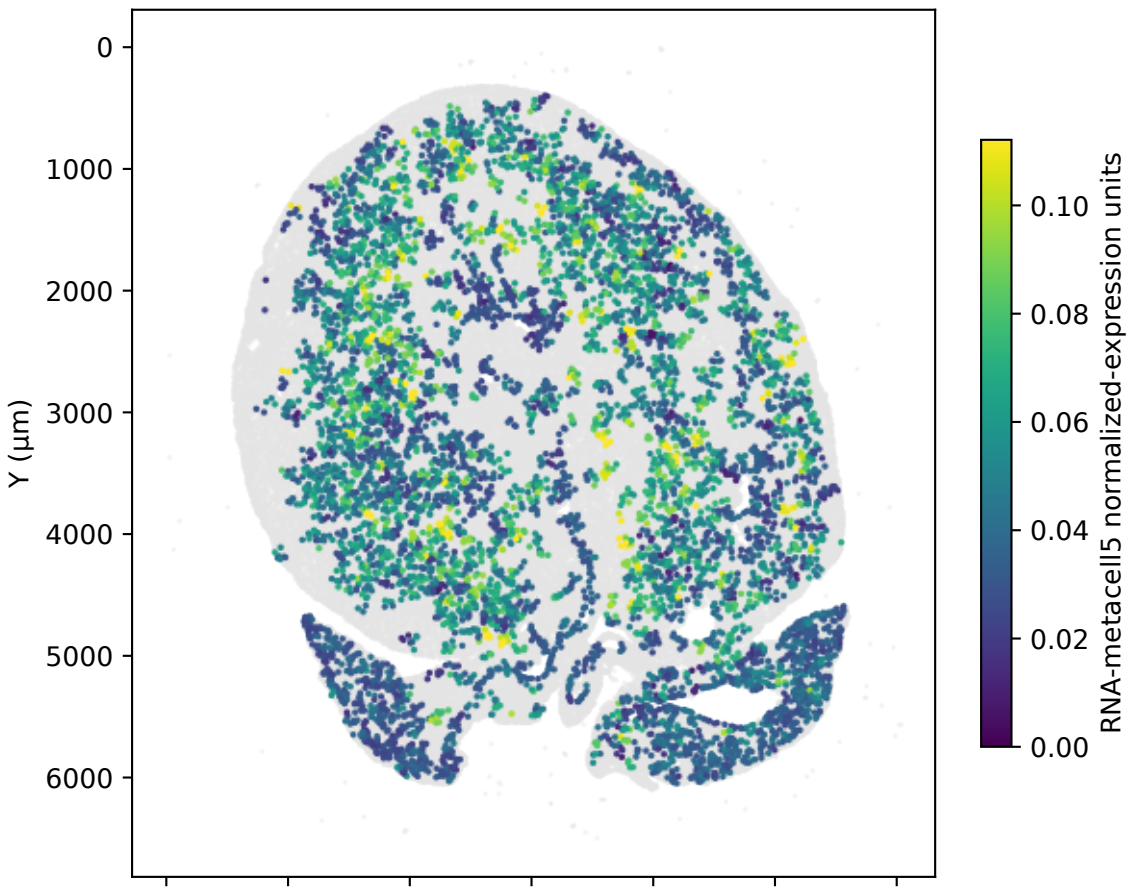
Five ECs within75µm or20 within150µm; overlapping pools, not independent cells; scales differ by RNA reference resolution



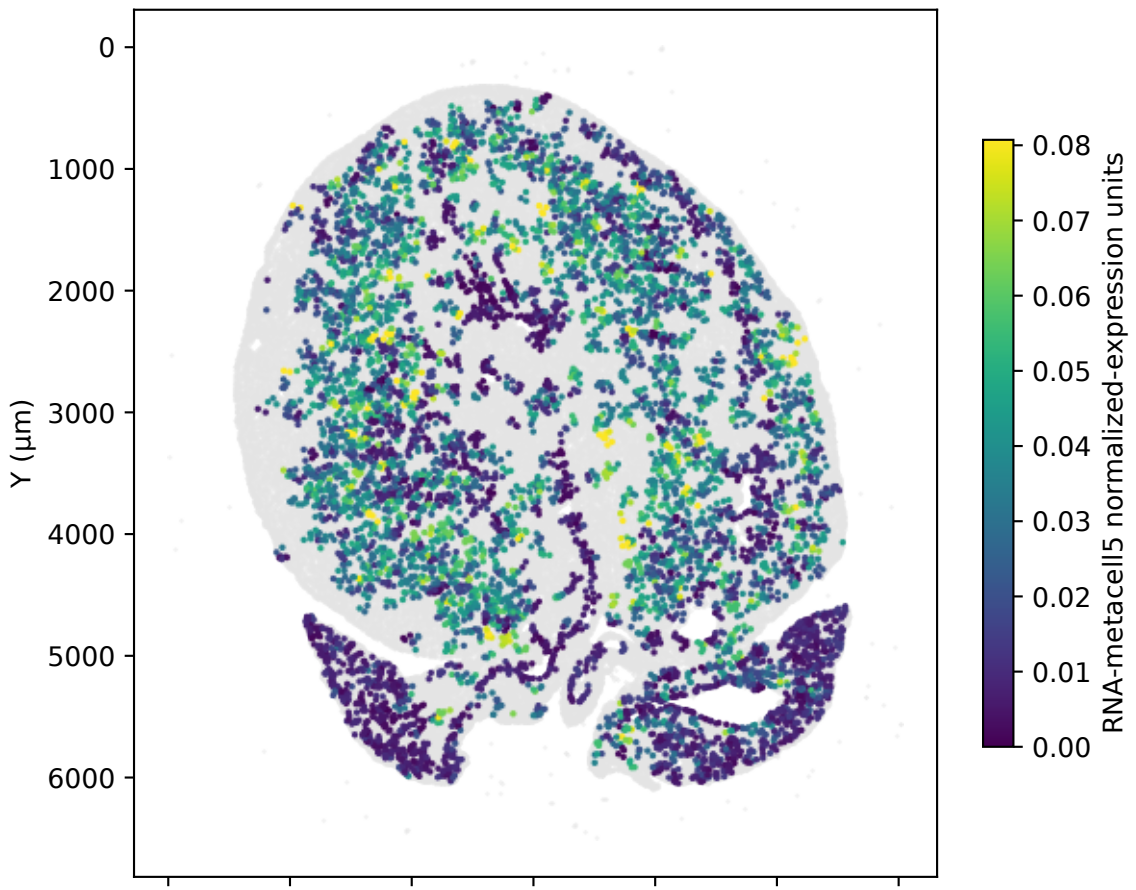
P21 · exploratory RNA-inferred neighborhood sprouting

Five ECs within75µm or20 within150µm; overlapping pools, not independent cells; scales differ by RNA reference resolution

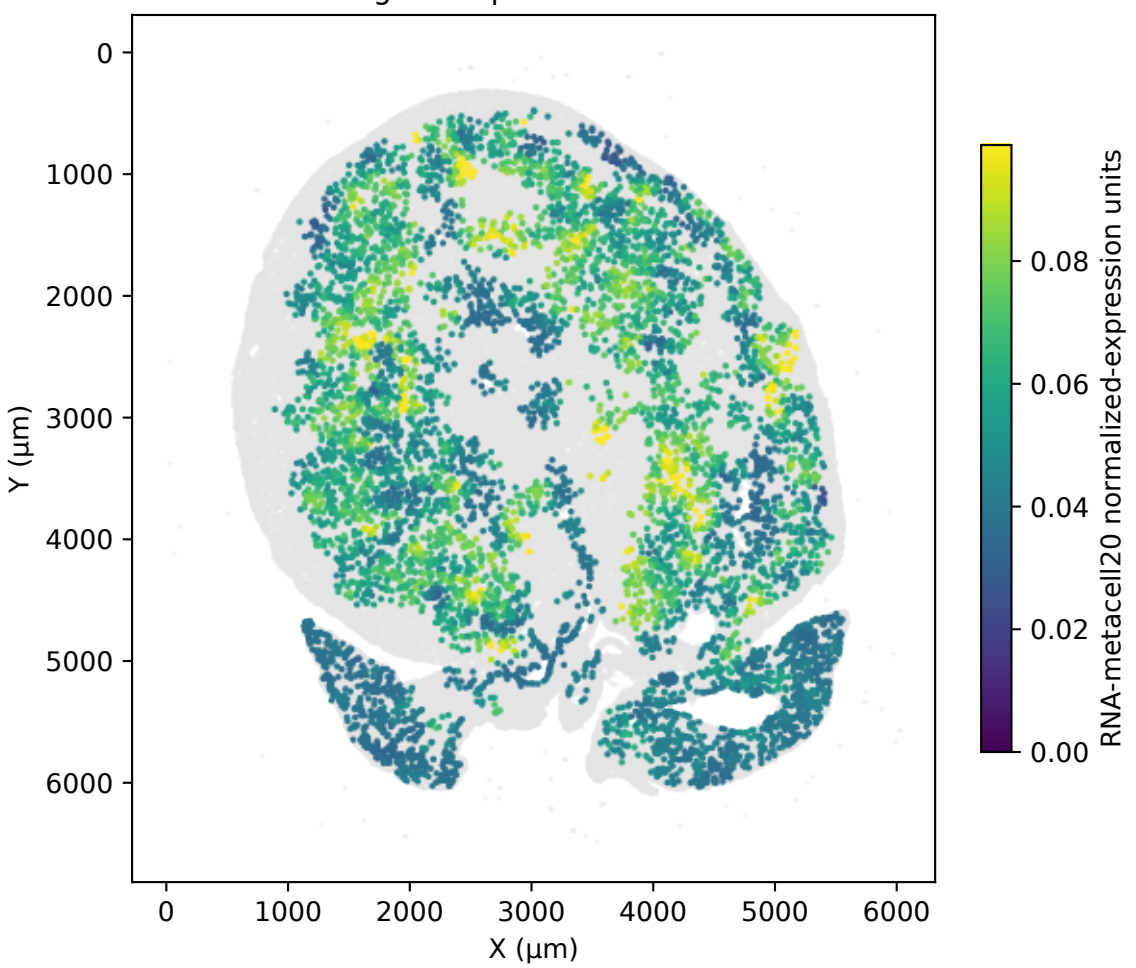
Six-gene expectation ·5 ECs



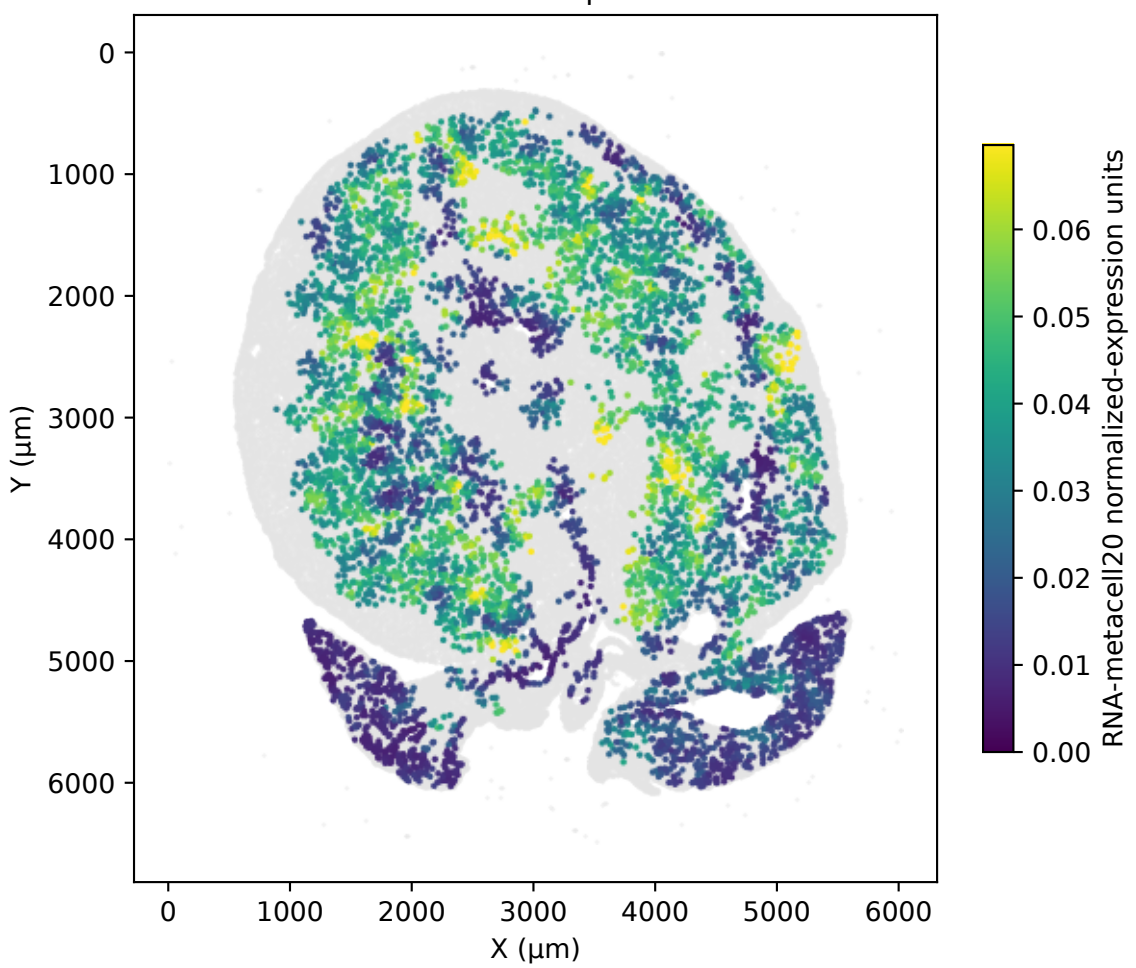
Unmeasured-five component ·5 ECs



Six-gene expectation ·20 ECs



Unmeasured-five component ·20 ECs



How to use the inferred layer

Each Xenium_ECs_with_inferred_sprouting.h5ad preserves observed native gene counts inX and403 observed gene features invar. Nuclear and boundary-core counts are retained as observed layers. Projected six-gene expression is stored separately inobsm, in RNA log1pCP10k units. Program scores and uncertainty diagnostics are inobs, with micrometer coordinates inobsm[spatial].

The whole-heart TSV includes all original cardiac positions and identity labels, with inferred values only where EC/nuclear/geometry eligibility was satisfied. Join onstage pluscell_id, or the supplied unique stage_cell_id. Do not interpret missing scores as biological absence or add inferred expression to measured counts.

Raw-reference,10/30/60-neighbor, native/nuclear/boundary-core and Angpt2-withheld sensitivities accompany the primary projection. Reference-neighbor heterogeneity is not a calibrated prediction interval. RNA-library differences and low reference similarity flag possible instability; they do not independently validate a score.

Technical validation uses the opposite RNA library at the same age, with no unmeasured target or cycling-control target among fitting genes. Angpt2 withholding excludes it even from predictor normalization. RNA metacells are disjoint groups formed only from measured features; final small remainders are merged with a nearby group. Spatial neighborhoods overlap and can span neighboring vessels.

Projection can add a useful reference-derived hypothesis layer to Xenium without adding actual molecules. The six-gene score is partly informed by directly measured Angpt2; confidence in the unmeasured five-gene component must be assessed separately. Sparse reference genes and age/library composition limit fine spatial inference. Independent targeted RNA in situ would be needed to verify unmeasured sprouting genes.